

Afeke College of Engineering, Tel-Aviv

Smart Thermal System for Patient Safety Monitoring

Human & Fire Detection — Algorithm Evaluation

Waveshare 26984 Thermal Sensor 80×62 px 17 Scenes 8,259 Annotated Frame-Views

Guy Chen

Yaniv Blau

Roy Lieberman

Supervisor: Or Zilberberg

Advisor: Dr. Oshrit Hoffer

June 2026

Contents

1	Overview	2
1.1	Sensor and dataset	2
1.2	Common methodology	2
Part I — Human Detection		3
2	Setup (Human Detection)	3
2.1	Scenes and split	3
2.2	Detectors	3
2.3	Success criteria	3
3	Results (Human Detection)	4
3.1	Aggregate (test split)	4
3.2	Reading the results	4
Part II — Fire Detection		5
4	Setup (Fire Detection)	5
4.1	Scenes and balance	5
4.2	Detectors	5
5	Results (Fire Detection)	5
5.1	Aggregate	5
5.2	Per-scene breakdown	6
5.3	The recall ceiling on cigarette scenes	6
6	Conclusions	6

1 Overview

This report evaluates the human-detection and fire-detection algorithms of the Smart Thermal System on the **new Waveshare 26984 dataset**: three thermal sensors at 80×62 px and 8 Hz, recorded across 17 scenes in the ward setting. It supersedes the earlier MLX90640 (32×24) evaluations and is organised in two parts:

Part I — Human Detection:

three detectors (AdaptiveThreshold, HOG+SVM, MobileNet-SSD) under a frame-level presence criterion and a strict $\text{IoU} \geq 0.5$ localisation criterion.

Part II — Fire Detection:

two detectors (Otsu pipelined rule-based, and Fire-SVM) scored against warm-body hard negatives.

The headline result is that the higher-resolution, lower-noise Waveshare sensor substantially improves every detector relative to the MLX90640 baseline. Fire recall in particular more than doubles, and both fire detectors hold **near-perfect precision** (≤ 1 false alarm across 6,886 fire-negative frames that include all multi-person scenes).

1.1 Sensor and dataset

Table 1: Sensor upgrade. The Waveshare provides $\approx 6.5\times$ more pixels and less than half the noise floor of the MLX90640.

	MLX90640 (old)	Waveshare 26984 (new)
Resolution	32×24 (768 px)	80×62 (4,960 px)
Noise floor	1.5°C	0.7°C
Frame rate	8 Hz	8 Hz
Annotated fire frames	78	1,373
Annotated person frames	$\sim 5,900$	7,421

1.2 Common methodology

Every frame is reduced to a binary verdict and scored with a confusion matrix and four derived metrics:

$$\begin{aligned} \text{Accuracy} &= \frac{TP+TN}{TP+TN+FP+FN}, & \text{Precision} &= \frac{TP}{TP+FP}, \\ \text{Recall} &= \frac{TP}{TP+FN}, & \text{F1} &= \frac{2\text{P}\cdot\text{R}}{\text{P}+\text{R}}. \end{aligned}$$

Recall is the primary safety metric (a missed fire or intruder is a direct failure); **precision** guards against alert fatigue. All labels follow the project convention: YOLO class 0 = fire / ignition source, class 1 = person. Preprocessing (where used) is the Tateno pipeline (Gaussian denoise $\rightarrow$ background subtraction $\rightarrow$ L1 rectification), calibrated on even-indexed empty-room frames.

Part I — Human Detection

2 Setup (Human Detection)

2.1 Scenes and split

The only truly empty recording is `empty_room` (0 person boxes). Its even-indexed frames calibrate one Tateno pipeline per channel; its odd-indexed frames become true negatives. Every other labeled scene contributes person-present positives. Within each (scene, channel) group the labeled frame indices are sorted and split sequentially 80/10/10 into train/val/test, which prevents temporal leakage while keeping every scene represented in every split.

Table 2: Human-detection split statistics (frame-channel examples). Negatives are empty frames: empty-room odd frames plus person-scene frames whose subject has left the field of view.

Split	Total	Person-present (Pos)	Empty (Neg)
Train	6,441	5,866	575
Validation	801	783	18
Test	825	772	53

The dataset is strongly positive-skewed ($\approx 94\%$ person-present), reflecting a ward in which people are present almost continuously; the small empty-frame count makes the false-alarm rate a higher-variance statistic than recall.

2.2 Detectors

1. **AdaptiveThreshold** — rule-based local-contrast segmentation on the Tateno residual; no training. Resolution-parameterised.
2. **HOG+SVM** — Histogram-of-Oriented-Gradients features with a linear SVM, trained on the train split. Resolution-parameterised.
3. **MobileNet-SSD** — a deep single-shot detector (PyTorch, CUDA / RTX 2060 Super), 30 epochs. Resolution-fixed (a dedicated Waveshare checkpoint).

2.3 Success criteria

Criterion A — Presence:

a frame counts as a positive prediction if the detector emits *any* box; for empty frames any box is a false alarm.

Criterion B — Localisation ($\text{IoU} \geq 0.5$):

a positive frame is a TP only if a predicted box overlaps a ground-truth box with $\text{IoU} \geq 0.5$.

The aggregate tables below report Criterion B (the strict criterion); the *mean IoU on positive frames* is reported alongside.

3 Results (Human Detection)

3.1 Aggregate (test split)

Table 3: Human detection — aggregate test-split metrics under the strict localisation criterion ($\text{IoU} \geq 0.5$). MobileNet-SSD dominates on every axis; the classical detectors keep competitive recall but raise far more false alarms on empty frames.

Detector	Acc	Prec	Rec	F1	Mean IoU	False-alarm
AdaptiveThreshold	76.6%	93.4%	80.7%	86.6%	56.5%	83.0%
HOG+SVM	79.8%	92.7%	85.1%	88.7%	58.5%	98.1%
MobileNet-SSD	95.8%	99.1%	96.4%	97.7%	67.3%	13.2%

Table 4: Human-detection confusion counts on the test split (772 person frames, 53 empty frames). **TP** = person detected, **FN** = person missed, **FP** = false alarm on empty, **TN** = correct silence.

Detector	TP	FN	FP	TN
AdaptiveThreshold	623	149	44	9
HOG+SVM	657	115	52	1
MobileNet-SSD	744	28	7	46

3.2 Reading the results

- **MobileNet-SSD is the production choice.** It detects 744 of 772 person frames (96.4 % recall), localises them well (67.3 % mean IoU), and false-alarms on only 7 of 53 empty frames. On the crowded scenes (3ppl_dance, 3pp_surprise, 3pplhedroncolider, 2ppl_fight) it scores a perfect 100 % F1.
- **The classical detectors trade precision for simplicity.** AdaptiveThreshold (no training) and HOG+SVM both keep recall in the 80–85 % range, but each emits a spurious box on nearly every *empty* frame — HOG fired on 52 of 53 (98 % false-alarm). With people present almost continuously this barely dents accuracy, but it makes them unsuitable as a standalone presence gate.
- **The shared hard case is 1_man_run.** Its test tail is mostly the runner leaving the frame, so it is dominated by empty frames with residual motion heat; every detector struggles (MobileNet 40 % recall, classical detectors < 22 % accuracy). This is the scene to target with more training data.
- **Resolution helps localisation.** Mean IoU on the 80×62 sensor reaches 67 % for MobileNet — a box that is genuinely useful for the downstream contact / restricted-area logic, which the 32×24 MLX could not reliably provide.

Part II — Fire Detection

4 Setup (Fire Detection)

4.1 Scenes and balance

Fire-positive scenes are those containing ≥ 1 class-0 (fire) box: `1person_cig`, `heater_in_middle`, `man_light_cig`, and `man_light_cig_1`. All remaining 13 scenes are fire-negative and act as **hard negatives**: they contain warm human bodies ($\sim 33^\circ\text{C}$) and a warm calibration room that the detector must not confuse with an ignition source. Across all three channels the dataset is **1,373 fire / 6,886 safe** frame-views (16.6% positive).

4.2 Detectors

1. **OtsuFireDetector** — the rule-based three-stage pipeline: absolute hot-pixel segmentation (hot-pixel bypass, $T_{\text{ign}} = 45^\circ\text{C}$, $T_{\text{fire}} = 60^\circ\text{C}$), a hierarchical area/temperature classifier, and a temporal mass-gradient confirmation stage. No training; evaluated on every frame.
2. **FireSVMDetector** — Otsu blob features fed to an RBF SVM, trained on a 70/30 stratified split (per scene $\times$ camera $\times$ fire/safe).

5 Results (Fire Detection)

5.1 Aggregate

Table 5: Fire detection — aggregate metrics. Precision is near-perfect for both detectors; the Fire-SVM lifts recall by ≈ 19 points over the rule-based pipeline at the cost of one false alarm.

Detector	Acc	Prec	Rec	F1	False-alarm	Frames
Otsu (rule-based)	92.8%	99.9%	56.9%	72.5%	0.01%	8,259 (all)
Fire-SVM (RBF, test)	95.9%	99.7%	75.5%	85.9%	0.05%	2,506 (30%)

Otsu confusion (all 8,259 frames).

	Pred: Fire	Pred: Safe
Truth: Fire (1,373 fr.)	TP = 781	FN = 592
Truth: Safe (6,886 fr.)	FP = 1	TN = 6,885

Fire-SVM confusion (test split, 2,506 frames).

	Pred: Fire	Pred: Safe
Truth: Fire (416 fr.)	TP = 314	FN = 102
Truth: Safe (2,090 fr.)	FP = 1	TN = 2,089

5.2 Per-scene breakdown

Table 6: Fire detection, per fire-positive scene. Recall tracks how hot the boxed object actually is: the lighter and heater scenes are detected reliably; the cigarette scenes are recall-limited (see §5.3).

Scene	Detector	Prec	Rec	F1	Mean IoU	Frames
man_light_cig_1	Otsu	100%	100%	100%	35.6%	285
	SVM	100%	100%	100%	–	88
heater_in_middle	Otsu	100%	76.8%	86.9%	21.2%	1,137
	SVM	100%	92.9%	96.3%	–	343
man_light_cig	Otsu	100%	32.6%	49.1%	3.9%	210
	SVM	100%	50.0%	66.7%	–	65
1person cig	Otsu	100%	26.4%	41.7%	4.6%	1,029
	SVM	100%	50.0%	66.7%	–	312

5.3 The recall ceiling on cigarette scenes

The two cigarette scenes are recall-limited for a physical, not algorithmic, reason. The annotator boxes the *ignition-source object* (the cigarette) in every frame it is visible, including the many frames where the tip is not glowing. Measured peak temperatures inside the fire boxes confirm this:

Table 7: Peak temperature inside fire boxes. The detector’s T_{ign} is 45 °C; a frame whose boxed object never exceeds it cannot be detected by any absolute-threshold rule, which bounds achievable recall.

Scene	Median peak	>45 °C	>50 °C	>60 °C
man_light_cig_1	63.4 °C	100%	92%	75%
heater_in_middle	54.5 °C	77%	62%	45%
man_light_cig	42.7 °C	33%	19%	0%
1person cig	39.9 °C	26%	19%	10%

Recall closely tracks the “>45 °C” column: where the boxed object is genuinely hot (`man_light_cig_1`, `heater_in_middle`) recall is high; where it is mostly a cold object (`1person cig`) recall is bounded near the fraction of hot frames. This is the same label-vs-physics ceiling identified on the MLX dataset — but the better sensor pushes aggregate recall from 9% (Otsu) / 32% (SVM) on MLX to **56.9%** / **75.5%** here.

6 Conclusions

- **Fire detection is precision-safe and recall-improved.** Both detectors raise essentially zero false alarms (1 FP total, on a warm calibration-room frame) while the Fire-SVM reaches 75.5% recall — more than double the MLX baseline. For sustained, genuinely hot sources (lighter, heater) recall is 93–100%.

- **Residual misses are concentrated on cold cigarette frames**, an annotation-vs-physics artifact rather than a detector failure. A cigarette whose tip is below 45 °C carries no thermal signature to detect.
- **Human and fire detection are complementary** in the runtime pipeline; the per-detector trade-offs in Part I inform which human detector to pair with the fire stage.